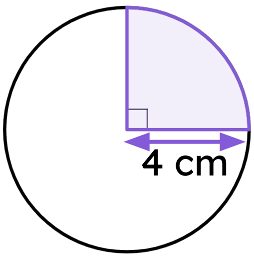
Name: ______________________

Worksheet

Perimeter of composite shapes

Task A: Comparing arcs and perimeters of a sector

1) The **arc** and sector are congruent to the one on the circle.
Complete each statement with both a calculation and an answer shown on the right.



The circumference of this circle is

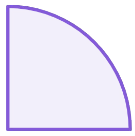
____

= ________

The **arc** length is

____

= ________



The perimeter of this sector is

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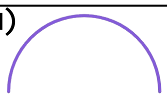
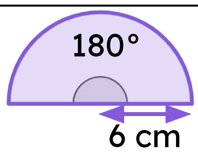
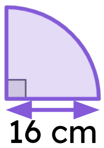
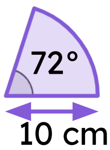
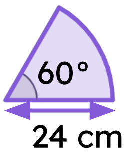
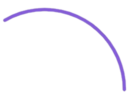
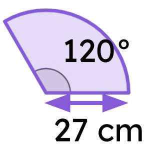
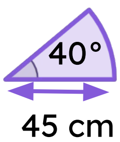
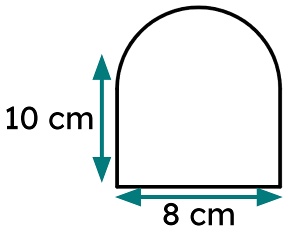
$2\pi + 8$	$4\pi + 4$
$\frac{1}{4} \times 8\pi$	$2\pi + 4$
$\frac{1}{4} \times 16\pi$	$2 \times \pi \times 4$
$\pi \times 4^2$	16π, 4π, 8π, 2π

2) Complete the table. Leave your answers in terms of π.

arc	corresponding sector	fraction of a full circle	calculations for **arc** length	calculations for perimeter of sector
a) 				$6\pi + 12$
b) 				
c) 		$\frac{72}{360} = \frac{1}{5}$	$\frac{1}{5} \times 2 \times \pi \times 10$ $= 4\pi$	$4\pi + 10 + 10$
Complete the table. Round your answers to 1 decimal place.				
d) 				73.1 cm
e) 			56.5 cm	
f) 				

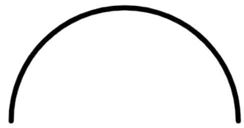
Task B: Finding perimeters of closed composite shapes

1) The composite shape has been broken down into its component lengths.



a) Write down the length of all three of these straight lines.

b) What is the radius of this semicircle?



c) Complete this calculation to find the **arc** length of this semicircle.

$$\frac{1}{2} \times 2 \times \pi \times \_\_\_\_ = \_\_\_\_$$

d) Hence, find the total perimeter of this shape.

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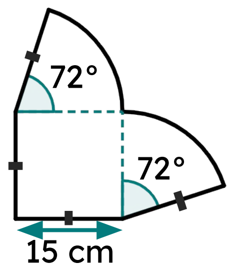
2) The composite shape has been broken down into its component lengths.



a) Write down the length of all four of these straight lines.

b) What fraction of a full circle do each of these **arcs** represent?

c) Complete this calculation to find the **arc** length of each **arc**.

$$\_\_\_ \times 2 \times \pi \times \_\_\_\_ = \_\_\_\_$$

d) Hence, find the total perimeter of this shape.

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3) Starting with the smallest perimeter, sort these shapes in terms of the size of their perimeters.